Federated Learning

General Review and Future Directions

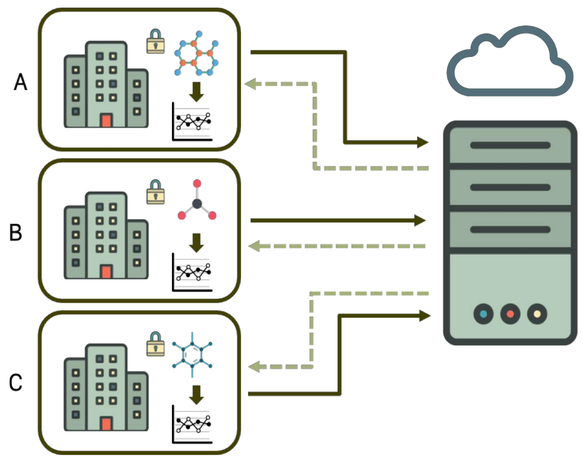
María Chavero-Díez[1,2]; Socayna Jouide[3,4]; Carles Hernandez-Ferrer[1,2]; Laia Codó[1,2]; Karim Lekadir [4] ; Salvador Capella-Gutierrez[1,2]; Josep Lluís Gelpí[1,2,3]

[1] Spanish National Bioinformatics Institute (INB/ELIXIR-ES)
[2] Barcelona Supercomputing Center (BSC)
[3] Dept Biochemistry and Molecular Biomedicine, University of Barcelona (UB)
[4] Dept Mathematics and Informatics, University of Barcelona (UB)

WHAT IS FEDERATED LEARNING?

Federated learning (FL) is defined as a decentralized machine learning technique that enables the training of learning models using data that is distributed across different devices or servers, without the need to centralize all the data in a single place.

In FL, the data is kept on local devices or servers, and learning algorithms are used that allow the exchange of information only in the form of model parameters and not the data itself This allows data to remain secure and private on the source devices.



WHY IS A REVIEW ON FEDERATED LEARNING NEEDED?

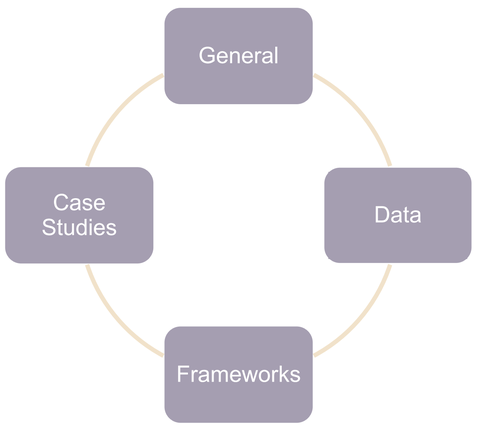
A review of the current state of the art regarding federated machine learning in the biomedical field could provide valuable and novel information on the strengths and limitations of this technology, as well as identify areas in which more research is needed.

This work also intends to serve as a starting point and roadmap for future developments that take place in projects associated with ELIXIR and ELIXIR nodes, such as the European Genomic Data Infrastructure (GDI) and the EUropean Federation for CAncer IMages (EUCAIM).

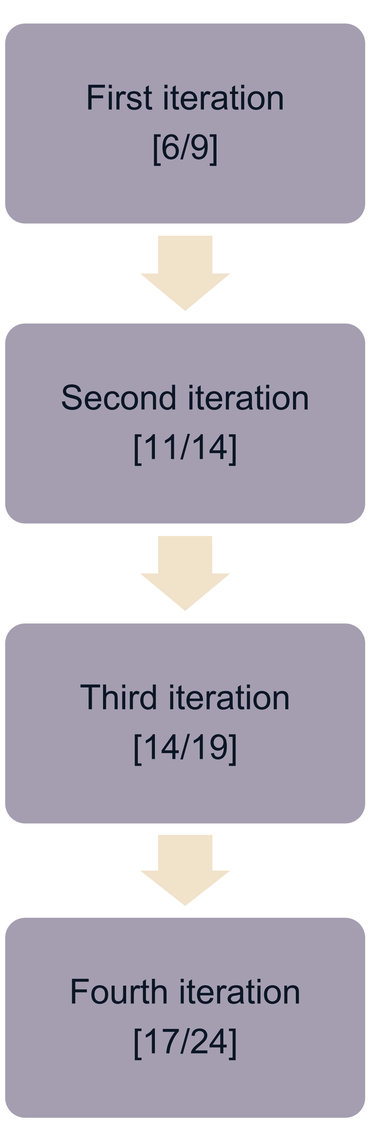
REVISION CRITERIA & FOCUS POINTS

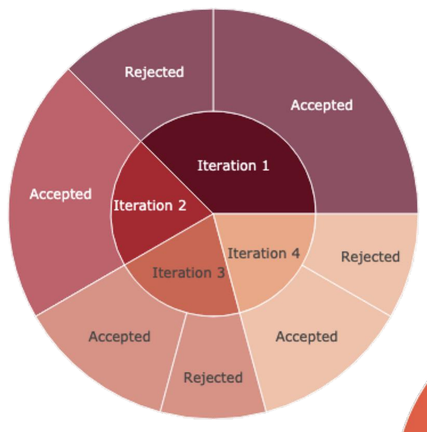
Within the framework of this bibliographical review, four main thematic pillars have been selected:

- The first pillar, called "General", covers those studies that offer a broad vision based on the general bibliography.
- The second pillar, titled "Data", groups the articles focused on the preservation and ownership of data.
- The third pillar, called "Frameworks", focuses on studies that propose tools that facilitate FL studies.
- Finally, the fourth pillar, entitled "Case Studies" brings together the articles that present specific use cases of federated analysis.

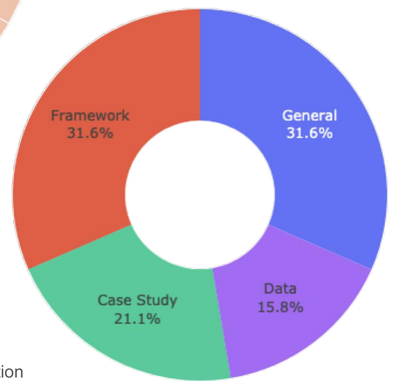


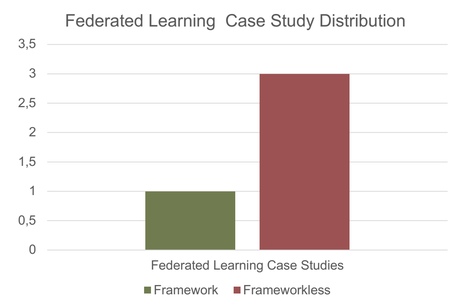
Review Process





Items of Study
Title
Abstract
Full text
Technology
Topic





(*) Number of scientific articles included acquired in each iteration and non-cumulative included in each category being, from left to right: General, Data, Framework, and Case Study.

DISCLAIMER!

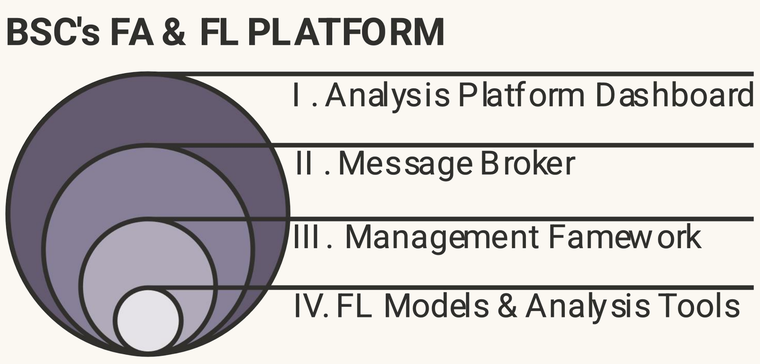
This is an ongoing work, still at an **early stage**. This means that, although, we are observing some interesting patterns, like the lack of case studies that are linked to existing FL frameworks, further study of the literature is needed in order to reach a consensus about the actual current state of the art and the issues that federated learning is facing.

HOW DOES THIS TRANSLATE INTO REAL LIFE? OUR FIRST PILOT IMPLEMENTATION

In our involvement in multiple Spanish and European projects, we were required to develop a system to make federated analysis and federated learning available in multiple scenarios. This led to the development of a multiple layers system, each layer dedicated to a specific goal. This modular architecture allows the exchange of the selected solution for any specific task, easily adapting the framework to each project's requirements:

- The high-level architecture of our Federated Processing system involves a central platform that coordinates the collaboration between the data nodes involved in the federated analysis. Then, it manages the results under the type of analyses performed (for instance: summarized for federate analysis and aggregation for federated learning).
- Our Federated Processing system is based on an engine that can deploy and support the entire life cycle of Federated Analysis (FA) or Federated Learning (FL) experiments.

As depicted in the figure, using a federated learning use-case as an example, our system follows a specific architecture:



General description of the Federated Processing system developed at BSC. It is composed of four main layers that can be exchanged to use other tools to fit multiple scenarios.

I. Analysis Platform Dashboard: web user interface where the user triggers an experiment and retrieves the results. In this dashboard, the user will configure the complete experiment by selecting variables, data nodes, and models to used.

II. Message broker: currently based on RabbitMQ, it is the responsible for the communications between all nodes. Including the communication between the central node and data nodes to collect models and results, but also the communication to orchestrate any experiment. The use of a message broker allows to operate in restricted networking scenarios.

III. Management framework: It is responsible to raise the dynamic infrastructure to trigger the experiment at each node. Our implementation, by default, uses a series of docker-based implementations of the Flower client (at the data nodes) and Flower server (at the central node).

IV. Federated Learning models: The selected model (for both federated analysis and federated learning) are materialised using git and mounted into the Flower client docker images at the data nodes.

In a more general case, Flower can be replaced by containers providing the desired server and client functionalities and/or adapted to be compatible with a different "management framework" implementations.

María Chavero Díez
maria.chavero@bsc.es
Barcelona Supercomputing Center (BSC)
Spanish National Bioinformatics Institute / ELIXIR-ES
c/Jordi Girona, 29; 08034 Barcelona; Spain

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